Highlights

- Brain-inspired 4-DoF localization-mapping framework for 3D spatial cognition
- Dual-stream visual processing with vestibular IMU fusion on a 10-state EKF core
- Scale and loop-closure corrections trained once in simulation, deployed zero-shot
- Outperforms the EKF baseline on all five scenarios under Sim(3) and SE(3) calibers
- Runs at 39–76 FPS on CPU (measured, no GPU) with an interpretable experience map